

1. Education Analytics Dataset

Question 1

A college wants to analyze student performance based on attendance, study hours, internal marks, and final result. Perform EDA to identify the factors affecting student results.

Student_ID	Attendance	Study_Hours	Internal_Marks	Result
S01	92	5	45	Pass
S02	55	2	24	Fail
S03	78	4	38	Pass
S04	60	3	28	Fail
S05	88	5	42	Pass
S06	47	1	20	Fail
S07	95	6	48	Pass
S08	72	3	35	Pass
S09	50	2	22	Fail
S10	83	4	40	Pass
S11	66	3	30	Fail
S12	90	5	44	Pass
S13	58	2	25	Fail
S14	76	4	36	Pass
S15	62	3	29	Fail
S16	85	5	41	Pass
S17	45	1	18	Fail
S18	80	4	39	Pass
S19	68	3	32	Pass
S20	53	2	23	Fail

Answer:

```
# =====  
# EDUCATION ANALYTICS - EDA  
# =====
```

```
# Import Libraries  
import pandas as pd  
import matplotlib.pyplot as plt
```

```
# Upload CSV File  
from google.colab import files  
uploaded = files.upload()
```

```
# Read Dataset  
df = pd.read_csv("Education_Analytics.csv")
```

```
# -----  
# Basic Information  
# -----
```

```
print("First 5 Records")  
print(df.head())
```

```

print("\nDataset Shape:")
print(df.shape)

print("\nColumns:")
print(df.columns)

print("\nData Types:")
print(df.dtypes)

print("\nMissing Values:")
print(df.isnull().sum())

print("\nDuplicate Values:")
print(df.duplicated().sum())

print("\nStatistical Summary:")
print(df.describe())

print("\nPass / Fail Count:")
print(df["Result"].value_counts())

# -----
# Correlation
# -----

numeric_df = df[["Attendance", "Study_Hours", "Internal_Marks"]]
corr = numeric_df.corr()

print("\nCorrelation Matrix")
print(corr)

# =====
# 1. Bar Chart
# =====

result_count = df["Result"].value_counts()

plt.figure(figsize=(5,4))
plt.bar(result_count.index, result_count.values)
plt.title("Pass vs Fail Students")
plt.xlabel("Result")
plt.ylabel("Number of Students")
plt.show()

# =====
# 2. Histogram
# =====

plt.figure(figsize=(6,4))
plt.hist(df["Attendance"], bins=5)

```

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plt.title("Attendance Distribution")
plt.xlabel("Attendance")
plt.ylabel("Frequency")
plt.show()

# =====
# 3. Scatter Plot
# =====

plt.figure(figsize=(6,4))
plt.scatter(df["Study_Hours"], df["Internal_Marks"])
plt.title("Study Hours vs Internal Marks")
plt.xlabel("Study Hours")
plt.ylabel("Internal Marks")
plt.show()

# =====
# 4. Heatmap
# =====

plt.figure(figsize=(5,4))
plt.imshow(corr, cmap="coolwarm")
plt.colorbar()

plt.xticks(range(len(corr.columns)), corr.columns)
plt.yticks(range(len(corr.columns)), corr.columns)

plt.title("Correlation Heatmap")
plt.show()

```

Output:

	Student_ID	Attendance	Study_Hours	Internal_Marks	Result
0	S01	92	5	45	Pass
1	S02	55	2	24	Fail
2	S03	78	4	38	Pass
3	S04	60	3	28	Fail
4	S05	88	5	42	Pass

Dataset Shape:
(20, 5)

Columns:
Index(['Student_ID', 'Attendance', 'Study_Hours', 'Internal_Marks', 'Result'], dtype='object')

Data Types:

Student_ID	object
Attendance	int64
Study_Hours	int64
Internal_Marks	int64

Result object
dtype: object

Missing Values:

Student_ID 0
Attendance 0
Study_Hours 0
Internal_Marks 0
Result 0
dtype: int64

Duplicate Values:

0

Statistical Summary:

	Attendance	Study_Hours	Internal_Marks
count	20.000000	20.000000	20.000000
mean	70.150000	3.350000	32.950000
std	15.938286	1.424411	9.087151
min	45.000000	1.000000	18.000000
25%	57.250000	2.000000	24.750000
50%	70.000000	3.000000	33.500000
75%	83.500000	4.250000	40.250000
max	95.000000	6.000000	48.000000

Pass / Fail Count:

Result

Pass 11

Fail 9

Name: count, dtype: int64

Correlation Matrix

	Attendance	Study_Hours	Internal_Marks
Attendance	1.000000	0.975888	0.997569
Study_Hours	0.975888	1.000000	0.977299
Internal_Marks	0.997569	0.977299	1.000000



